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BICYCLIC PEPTIDE LIGANDS SPECIFIC FOR TRANSFERRIN RECEPTOR 1 (TfR1)

Abstract

The present invention relates to peptide ligands, such as bicyclic peptide ligands, specific for transferrin receptor 1 (TfR1). The invention also includes pharmaceutical compositions comprising said peptide ligands and the use of said peptide ligands and pharmaceutical compositions in preventing, suppressing or treating a disease or disorder through TfR1 mediated delivery of a therapeutic agent.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to peptide ligands, such as bicyclic peptide ligands, specific for transferrin receptor 1 (TfR1). The invention also includes pharmaceutical compositions comprising said peptide ligands and the use of said peptide ligands and pharmaceutical compositions in preventing, suppressing or treating a disease or disorder through TfR1 mediated delivery of a therapeutic agent.

BACKGROUND OF THE INVENTION

[0002] Cyclic peptides are able to bind with high affinity and specificity to protein targets and hence are an attractive molecule class for the development of therapeutics. In fact, several cyclic peptides are already successfully used in the clinic, as for example the antibacterial peptide vancomycin, the immunosuppressant drug cyclosporine or the anti-cancer drug octreotide (Driggers et al. (2008), Nat. Rev. Drug. Discov. 7(7), 608-24). Good binding properties result from a relatively large interaction surface formed between the peptide and the target as well as the reduced conformational flexibility of the cyclic structures. Typically, macrocycles bind to surfaces of several hundred square angstrom, as for example the cyclic peptide CXCR4 antagonist CVX15 (400 Å^{sup.2}; Wu et al. (2007), Science 330, 1066-71), a cyclic peptide with the Arg-Gly-Asp motif binding to integrin αVβ3 (355 Å^{sup.2}) (Xiong et al. (2002), Science 296(5565), 151-5) or the cyclic peptide inhibitor upain-1 binding to urokinase-type plasminogen activator (603 Å^{sup.2}; Zhao et al. (2007), J. Struct. Biol. 160(1), 1-10).

[0003] Due to their cyclic configuration, peptide macrocycles are less flexible than linear peptides, leading to a smaller loss of entropy upon binding to targets and resulting in a higher binding affinity. The reduced flexibility also leads to locking target-specific conformations, increasing binding specificity compared to linear peptides. This effect has been exemplified by a potent and selective inhibitor of matrix metalloproteinase 8 (MMP-8) which lost its selectivity over other MMPs when its ring was opened (Cherney et al. (1998), J. Med. Chem. 41(11), 1749-51). The favourable binding properties achieved through macrocyclization are even more pronounced in multicyclic peptides having more than one peptide ring as for example in vancomycin, nisin and actinomycin.

[0004] Different research teams have previously tethered polypeptides with cysteine residues to a synthetic molecular structure (Kemp and McNamara (1985), J. Org. Chem; Timmerman et al. (2005), ChemBioChem). Meloen and co-workers had used tris(bromomethyl)benzene and related molecules for rapid and quantitative cyclisation of multiple peptide loops onto synthetic scaffolds for structural mimicry of protein surfaces (Timmerman et al. (2005), ChemBioChem). Methods for the generation of candidate drug compounds wherein said compounds are generated by linking cysteine containing polypeptides to a molecular scaffold as for example 1,1',1''-(1,3,5-triazine-1,3,5-triyl)triprop-2-en-1-one (TATA) (Heinis et al. (2014) Angewandte Chemie, International Edition 53(6) 1602-1606).

[0005] Phage display-based combinatorial approaches have been developed to generate and screen large libraries of bicyclic peptides to targets of interest (Heinis et al. (2009), Nat. Chem. Biol. 5(7), 502-7 and WO 2009/098450). Briefly, combinatorial libraries of linear peptides containing three cysteine residues and two regions of six random amino acids (Cys-(Xaa).sub.6-Cys-(Xaa).sub.6-Cys) were displayed on phage and cyclised by covalently linking the cysteine side chains to a small molecule scaffold.

SUMMARY OF THE INVENTION

[0006] According to a first aspect of the invention, there is provided a peptide ligand specific for transferrin receptor 1 (TfR1) which comprises an amino acid sequence which is selected from: [0007] C[HyP]

[HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 1, herein referred to as BCY23180); [0008] C[Cis-HyP]

[HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 3, herein referred to as BCY23182); [0009] CP[Cis-

HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 5, herein referred to as BCY23184); [0010]

CP[HyP]DA[DOPA]LGC[tBuGly]SYCEPW (SEQ ID NO: 6, herein referred to as BCY23185); [0011]

CP[HyP]DA[pCaPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 7, herein referred to as BCY23186); [0012]

CP[HyP]DA[pCoPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 8, herein referred to as BCY23187); [0013]

CP[HyP]DA[hTyr]LGC[tBuGly]SYCEPW (SEQ ID NO: 9, herein referred to as BCY23188); [0014]

CP[HyP]DAYLGC[tBuGly]S[DOPA]CEPW (SEQ ID NO: 21, herein referred to as BCY23200); [0015]

CP[HyP]DAYLGC[tBuGly]S[pCaPhe]CEPW (SEQ ID NO: 22, herein referred to as BCY23201); [0016]

CP[HyP]DAYLGC[tBuGly]S[pCoPhe]CEPW (SEQ ID NO: 23, herein referred to as BCY23202); [0017]

CP[HyP]DAYLGC[tBuGly]S[hTyr]CEPW (SEQ ID NO: 24, herein referred to as BCY23203); [0018]

CP[HyP]DAYLGC[tBuGly]SYCE[HyP]W (SEQ ID NO: 25, herein referred to as BCY23204); [0019]

CP[HyP]DAYLGC[tBuGly]SYCE[Oxa]W (SEQ ID NO: 26, herein referred to as BCY23205); [0020]

CP[HyP]DAYLGC[tBuGly]SYCE[Cis-HyP]W (SEQ ID NO: 27, herein referred to as BCY23206); [0021]

CP[HyP]DAYLGC[tBuGly]SYCEPY (SEQ ID NO: 28, herein referred to as BCY23207); [0022]

CP[HyP]DAYLGC[tBuGly]SYCEP[DOPA](SEQ ID NO: 29, herein referred to as BCY23208); [0023]

CP[HyP]DAYLGC[tBuGly]SYCEP[pCaPhe](SEQ ID NO: 30, herein referred to as BCY23209); [0024]

CP[HyP]DAYLGC[tBuGly]SYCEP[pCoPhe](SEQ ID NO: 31, herein referred to as BCY23210); [0025]

CP[HyP]DAYLGC[tBuGly]SYCEP[hTyr](SEQ ID NO: 32, herein referred to as BCY23211); [0026]

CP[HyP]EAYLGC[tBuGly]SYCEPW (SEQ ID NO: 33, herein referred to as BCY23216); [0027] CP[HyP]

[Gla]AYLGC[tBuGly]SYCEPW (SEQ ID NO: 34, herein referred to as BCY23217); [0028]

CP[HyP]DAYSGC[tBuGly]SYCEPW (SEQ ID NO: 35, herein referred to as BCY23218); [0029]

CP[HyP]DAYTGC[tBuGly]SYCEPW (SEQ ID NO: 36, herein referred to as BCY23219); [0030]

CP[HyP]DAYDGC[tBuGly]SYCEPW (SEQ ID NO: 37, herein referred to as BCY23220); [0031]

CP[HyP]DAYEGC[tBuGly]SYCEPW (SEQ ID NO: 38, herein referred to as BCY23221); [0032]

CP[HyP]DAYNGC[tBuGly]SYCEPW (SEQ ID NO: 39, herein referred to as BCY23222); [0033]

CP[HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 40, herein referred to as BCY23223); [0034]
CP[HyP]DAYLGC[tBuGly][HSer]YCEPW (SEQ ID NO: 41, herein referred to as BCY23224); [0035]
CP[HyP]DAYLGC[tBuGly]SYCDPW (SEQ ID NO: 47, herein referred to as BCY23230); [0036]
CP[HyP]DAYLGC[tBuGly]SYC[Gla]PW (SEQ ID NO: 48, herein referred to as BCY23231); and [0037]
CP[HyP]DAYLGC[3HyV]SYCEPW (SEQ ID NO: 50, herein referred to as BCY23515), [0038] wherein Cis-HyP represents cis-L-4-hydroxyproline, DOPA represents 3,4-dihydroxy-phenylalanine, Gla represents L-γ-carboxyglutamic acid, HyP represents hydroxyproline, HSer represents homoserine, hTyr represents homo-tyrosine, 3HyV represents 3-hydroxy-L-valine, Oxa represents oxazolidine-4-carboxylic acid, pCaPhe represents L-4-carbamoylphenylalanine, pCoPhe represents 4-carboxy-L-phenylalanine, tBuGly represents t-butyl-glycine. [0039] According to a further aspect of the invention, there is provided a bicyclic peptide ligand which comprises a peptide ligand as defined herein wherein the first, second and third cysteine residues within said peptide ligands are covalently bonded to a molecular scaffold such that two polypeptide loops are formed on said molecular scaffold. [0040] According to a yet further aspect of the invention, there is provided a pharmaceutical composition comprising a peptide ligand or bicyclic peptide ligand as defined herein in combination with one or more pharmaceutically acceptable excipients. [0041] According to a further aspect of the invention, there is provided a peptide ligand, bicyclic peptide ligand or pharmaceutical composition as defined herein for use in preventing, suppressing or treating a disease or disorder through TfR1 mediated delivery of a therapeutic agent.

Description

DETAILED DESCRIPTION OF THE INVENTION

Peptide Ligands

[0042] According to a first aspect of the invention, there is provided a peptide ligand specific for transferrin receptor 1 (TfR1) which comprises an amino acid sequence which is selected from: [0043] C[HyP]
[HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 1, herein referred to as BCY23180); [0044] C[Cis-HyP]
[HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 3, herein referred to as BCY23182); [0045] CP[Cis-HyP]
DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 5, herein referred to as BCY23184); [0046]
CP[HyP]DA[DOPA]LGC[tBuGly]SYCEPW (SEQ ID NO: 6, herein referred to as BCY23185); [0047]
CP[HyP]DA[pCaPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 7, herein referred to as BCY23186); [0048]
CP[HyP]DA[pCoPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 8, herein referred to as BCY23187); [0049]
CP[HyP]DA[hTyr]LGC[tBuGly]SYCEPW (SEQ ID NO: 9, herein referred to as BCY23188); [0050]
CP[HyP]DAYLGC[tBuGly]S[DOPA]CEPW (SEQ ID NO: 21, herein referred to as BCY23200); [0051]
CP[HyP]DAYLGC[tBuGly]S[pCaPhe]CEPW (SEQ ID NO: 22, herein referred to as BCY23201); [0052]
CP[HyP]DAYLGC[tBuGly]S[pCoPhe]CEPW (SEQ ID NO: 23, herein referred to as BCY23202); [0053]
CP[HyP]DAYLGC[tBuGly]S[hTyr]CEPW (SEQ ID NO: 24, herein referred to as BCY23203); [0054]
CP[HyP]DAYLGC[tBuGly]SYCE[HyP]W (SEQ ID NO: 25, herein referred to as BCY23204); [0055]
CP[HyP]DAYLGC[tBuGly]SYCE[Oxa]W (SEQ ID NO: 26, herein referred to as BCY23205); [0056]
CP[HyP]DAYLGC[tBuGly]SYCE[Cis-HyP]W (SEQ ID NO: 27, herein referred to as BCY23206); [0057]
CP[HyP]DAYLGC[tBuGly]SYCEPY (SEQ ID NO: 28, herein referred to as BCY23207); [0058]
CP[HyP]DAYLGC[tBuGly]SYCEP[DOPA](SEQ ID NO: 29, herein referred to as BCY23208); [0059]
CP[HyP]DAYLGC[tBuGly]SYCEP[pCaPhe](SEQ ID NO: 30, herein referred to as BCY23209); [0060]
CP[HyP]DAYLGC[tBuGly]SYCEP[pCoPhe](SEQ ID NO: 31, herein referred to as BCY23210); [0061]
CP[HyP]DAYLGC[tBuGly]SYCEP[hTyr](SEQ ID NO: 32, herein referred to as BCY23211); [0062]
CP[HyP]EAYLGC[tBuGly]SYCEPW (SEQ ID NO: 33, herein referred to as BCY23216); [0063] CP[HyP]
[Gla]AYLGC[tBuGly]SYCEPW (SEQ ID NO: 34, herein referred to as BCY23217); [0064]
CP[HyP]DAYSGC[tBuGly]SYCEPW (SEQ ID NO: 35, herein referred to as BCY23218); [0065]
CP[HyP]DAYTGC[tBuGly]SYCEPW (SEQ ID NO: 36, herein referred to as BCY23219); [0066]
CP[HyP]DAYDGC[tBuGly]SYCEPW (SEQ ID NO: 37, herein referred to as BCY23220); [0067]
CP[HyP]DAYEGC[tBuGly]SYCEPW (SEQ ID NO: 38, herein referred to as BCY23221); [0068]
CP[HyP]DAYNGC[tBuGly]SYCEPW (SEQ ID NO: 39, herein referred to as BCY23222); [0069]
CP[HyP]DAYQGC[tBuGly]SYCEPW (SEQ ID NO: 40, herein referred to as BCY23223); [0070]
CP[HyP]DAYLGC[tBuGly][HSer]YCEPW (SEQ ID NO: 41, herein referred to as BCY23224); [0071]
CP[HyP]DAYLGC[tBuGly]SYCDPW (SEQ ID NO: 47, herein referred to as BCY23230); [0072]
CP[HyP]DAYLGC[tBuGly]SYC[Gla]PW (SEQ ID NO: 48, herein referred to as BCY23231); and [0073]
CP[HyP]DAYLGC[3HyV]SYCEPW (SEQ ID NO: 50, herein referred to as BCY23515), [0074] or a pharmaceutically acceptable salt of said peptide ligand thereof, wherein Cis-HyP represents cis-L-4-hydroxyproline, DOPA represents 3,4-dihydroxy-phenylalanine, Gla represents L-γ-carboxyglutamic acid, HyP represents hydroxyproline, HSer represents homoserine, hTyr represents homo-tyrosine, 3HyV represents 3-hydroxy-L-valine,

Oxa represents oxazolidine-4-carboxylic acid, pCaPhe represents L-4-carbamoylphenylalanine, pCoPhe represents 4-carboxy-L-phenylalanine, tBuGly represents t-butyl-glycine.

[0075] In a further aspect of the invention which may be mentioned, there is provided a peptide ligand specific for transferrin receptor 1 (TfR1) which comprises an amino acid sequence which is selected from: [0076] C[HyP][HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 1, herein referred to as BCY23180); [0077] C[Oxa][HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 2, herein referred to as BCY23181); [0078] C[Cis-HyP][HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 3, herein referred to as BCY23182); [0079]

CP[Oxa]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 4, herein referred to as BCY23183); [0080] CP[Cis-HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 5, herein referred to as BCY23184); [0081]

CP[HyP]DA[DOPA]LGC[tBuGly]SYCEPW (SEQ ID NO: 6, herein referred to as BCY23185); [0082]

CP[HyP]DA[pCaPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 7, herein referred to as BCY23186); [0083]

CP[HyP]DA[pCoPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 8, herein referred to as BCY23187); [0084]

CP[HyP]DA[hTyr]LGC[tBuGly]SYCEPW (SEQ ID NO: 9, herein referred to as BCY23188); [0085]

CP[HyP]DAYL[dS]C[tBuGly]SYCEPW (SEQ ID NO: 10, herein referred to as BCY23189); [0086]

CP[HyP]DAYL[dT]C[tBuGly]SYCEPW (SEQ ID NO: 11, herein referred to as BCY23190); [0087]

CP[HyP]DAYL[dD]C[tBuGly]SYCEPW (SEQ ID NO: 12, herein referred to as BCY23191); [0088]

CP[HyP]DAYL[dE]C[tBuGly]SYCEPW (SEQ ID NO: 13, herein referred to as BCY23192); [0089]

CP[HyP]DAYL[dN]C[tBuGly]SYCEPW (SEQ ID NO: 14, herein referred to as BCY23193); [0090]

CP[HyP]DAYL[dQ]C[tBuGly]SYCEPW (SEQ ID NO: 15, herein referred to as BCY23194); [0091]

CP[HyP]DAYL[dY]C[tBuGly]SYCEPW (SEQ ID NO: 16, herein referred to as BCY23195); [0092]

CP[HyP]DAYLSC[tBuGly]SYCEPW (SEQ ID NO: 17, herein referred to as BCY23196); [0093]

CP[HyP]DAYLDC[tBuGly]SYCEPW (SEQ ID NO: 18, herein referred to as BCY23197); [0094]

CP[HyP]DAYLYC[tBuGly]SYCEPW (SEQ ID NO: 19, herein referred to as BCY23198); [0095]

CP[HyP]DAYLNC[tBuGly]SYCEPW (SEQ ID NO: 20, herein referred to as BCY23199); [0096]

CP[HyP]DAYLGC[tBuGly]S[DOPA]CEPW (SEQ ID NO: 21, herein referred to as BCY23200); [0097]

CP[HyP]DAYLGC[tBuGly]S[pCaPhe]CEPW (SEQ ID NO: 22, herein referred to as BCY23201); [0098]

CP[HyP]DAYLGC[tBuGly]S[pCoPhe]CEPW (SEQ ID NO: 23, herein referred to as BCY23202); [0099]

CP[HyP]DAYLGC[tBuGly]S[hTyr]CEPW (SEQ ID NO: 24, herein referred to as BCY23203); [0100]

CP[HyP]DAYLGC[tBuGly]SYCE[HyP]W (SEQ ID NO: 25, herein referred to as BCY23204); [0101]

CP[HyP]DAYLGC[tBuGly]SYCE[Oxa]W (SEQ ID NO: 26, herein referred to as BCY23205); [0102]

CP[HyP]DAYLGC[tBuGly]SYCE[Cis-HyP]W (SEQ ID NO: 27, herein referred to as BCY23206); [0103]

CP[HyP]DAYLGC[tBuGly]SYCEPY (SEQ ID NO: 28, herein referred to as BCY23207); [0104]

CP[HyP]DAYLGC[tBuGly]SYCEP[DOPA](SEQ ID NO: 29, herein referred to as BCY23208); [0105]

CP[HyP]DAYLGC[tBuGly]SYCEP[pCaPhe](SEQ ID NO: 30, herein referred to as BCY23209); [0106]

CP[HyP]DAYLGC[tBuGly]SYCEP[pCoPhe](SEQ ID NO: 31, herein referred to as BCY23210); [0107]

CP[HyP]DAYLGC[tBuGly]SYCEP[hTyr](SEQ ID NO: 32, herein referred to as BCY23211); [0108]

CP[HyP]EAYLGC[tBuGly]SYCEPW (SEQ ID NO: 33, herein referred to as BCY23216); [0109] CP[HyP]

[Gla]AYLGC[tBuGly]SYCEPW (SEQ ID NO: 34, herein referred to as BCY23217); [0110]

CP[HyP]DAYSGC[tBuGly]SYCEPW (SEQ ID NO: 35, herein referred to as BCY23218); [0111]

CP[HyP]DAYTGC[tBuGly]SYCEPW (SEQ ID NO: 36, herein referred to as BCY23219); [0112]

CP[HyP]DAYDGC[tBuGly]SYCEPW (SEQ ID NO: 37, herein referred to as BCY23220); [0113]

CP[HyP]DAYEGC[tBuGly]SYCEPW (SEQ ID NO: 38, herein referred to as BCY23221); [0114]

CP[HyP]DAYNGC[tBuGly]SYCEPW (SEQ ID NO: 39, herein referred to as BCY23222); [0115]

CP[HyP]DAYQGC[tBuGly]SYCEPW (SEQ ID NO: 40, herein referred to as BCY23223); [0116]

CP[HyP]DAYLGC[tBuGly][HSer]YCEPW (SEQ ID NO: 41, herein referred to as BCY23224); [0117]

CP[HyP]DAYLGC[tBuGly]TYCEPW (SEQ ID NO: 42, herein referred to as BCY23225); [0118]

CP[HyP]DAYLGC[tBuGly]DYCEPW (SEQ ID NO: 43, herein referred to as BCY23226); [0119]

CP[HyP]DAYLGC[tBuGly]EYCEPW (SEQ ID NO: 44, herein referred to as BCY23227); [0120]

CP[HyP]DAYLGC[tBuGly]NYCEPW (SEQ ID NO: 45, herein referred to as BCY23228); [0121]

CP[HyP]DAYLGC[tBuGly]QYCEPW (SEQ ID NO: 46, herein referred to as BCY23229); [0122]

CP[HyP]DAYLGC[tBuGly]SYCDPW (SEQ ID NO: 47, herein referred to as BCY23230); [0123]

CP[HyP]DAYLGC[tBuGly]SYC[Gla]PW (SEQ ID NO: 48, herein referred to as BCY23231); [0124]

CP[HyP]DAYLGCYSYCEPW (SEQ ID NO: 49, herein referred to as BCY23514); and [0125]

CP[HyP]DAYLGC[3HyV]SYCEPW (SEQ ID NO: 50, herein referred to as BCY23515), [0126] wherein Cis-HyP

represents cis-L-4-hydroxyproline, DOPA represents 3,4-dihydroxy-phenylalanine, Gla represents L-γ-carboxyglutamic acid, HyP represents hydroxyproline, HSer represents homoserine, hTyr represents homo-tyrosine, 3HyV represents 3-hydroxy-L-valine, Oxa represents oxazolidine-4-carboxylic acid, pCaPhe represents L-4-carbamoylphenylalanine, pCoPhe represents 4-carboxy-L-phenylalanine, tBuGly represents t-butyl-glycine.

[0127] It will be appreciated that each of the peptide ligands of the invention comprise an N-terminal acetyl group

and a C-terminal CONH.sub.2 group.

[0128] It will also be appreciated that the term “specific for TfR1” refers to the ability of the peptide ligand to bind to transferrin receptor 1 (TfR1). It will also be appreciated that the peptide ligand will have a differing affect upon TfR1 depending on the precise epitope of binding. For example, the affect will either be inhibitory (i.e. the peptide ligand impedes/inhibits the binding of transferrin to TfR1) or non-inhibitory (i.e. the peptide ligand does not impede/inhibit the binding of transferrin to TfR1).

[0129] In a further embodiment, the pharmaceutically acceptable salt is selected from the free acid or the sodium, potassium, calcium or ammonium salt.

Bicyclic Peptide Ligands

[0130] According to a further aspect of the invention, there is provided a bicyclic peptide ligand which comprises a peptide ligand as defined herein wherein the first, second and third cysteine residues within said peptide ligands are covalently bonded to a molecular scaffold such that two polypeptide loops are formed on said molecular scaffold.

[0131] In one embodiment, the molecular scaffold is a derivative of TATB which has the following structure:

##STR00001## [0132] wherein * denotes the point of attachment of the three cysteine residues.

[0133] For the purpose of this description, bicyclic peptides are assumed to be cyclised with TATB to yield a tri-substituted structure. However, as will be clear from the descriptions of the invention presented herein, cyclisation may be performed with any suitable molecular scaffold which forms covalent bonds with the reactive groups of the polypeptide such that at least two polypeptide loops are formed. Cyclisation occurs on the first, second and third cysteine residues, respectively.

[0134] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art, such as in the arts of peptide chemistry, cell culture and phage display, nucleic acid chemistry and biochemistry. Standard techniques are used for molecular biology, genetic and biochemical methods (see Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 3rd ed., 2001, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY; Ausubel et al., *Short Protocols in Molecular Biology* (1999) 4.sup.th ed., John Wiley & Sons, Inc.), which are incorporated herein by reference.

Numbering

[0135] When referring to amino acid residue positions within the peptides of the invention, cysteine residues are omitted from the numbering as they are invariant, therefore, the numbering of amino acid residues within the peptides of the invention is referred to as below:

TABLE-US-00001 (SEQ ID NO: 1)

C[HyP].sub.1[HyP].sub.2D.sub.3A.sub.4Y.sub.5L.sub.6G.sub.7C[tBuGly].sub.8S.sub.9Y.sub.10CE.sub.11P.sub.12W.sub.13

Molecular Format

[0136] N- or C-terminal extensions to the bicycle core sequence are added to the left or right side of the sequence, separated by a hyphen. For example, an N-terminal biotin-G-Sar.sub.5 tail would be denoted as:

TABLE-US-00002 (SEQ ID NO: X) [Biot]-G-[Sar.sub.5]-A-.

Inversed Peptide Sequences

[0137] In light of the disclosure in Nair et al. (2003) *J. Immunol.* 170(3), 1362-1373, it is envisaged that the peptide sequences disclosed herein would also find utility in their retro-inverso form. For example, the sequence is reversed (i.e. N-terminus become C-terminus and vice versa) and their stereochemistry is likewise also reversed (i.e. D-amino acids become L-amino acids and vice versa).

Peptide Ligand Definition

[0138] A peptide ligand, as referred to herein, refers to a peptide, peptidic or peptidomimetic covalently bound to a molecular scaffold. Typically, such peptides, peptidics or peptidomimetics comprise a peptide having natural or non-natural amino acids, two or more reactive groups (i.e. cysteine residues) which are capable of forming covalent bonds to the scaffold, and a sequence subtended between said reactive groups which is referred to as the loop sequence, since it forms a loop when the peptide, peptidic or peptidomimetic is bound to the scaffold. In the present case, the peptides, peptidics or peptidomimetics comprise at least three cysteine residues, and form at least two loops on the scaffold.

Advantages of the Peptide Ligands

[0139] Certain bicyclic peptides of the present invention have a number of advantageous properties which enable them to be considered as suitable drug-like molecules for injection, inhalation, nasal, ocular, oral or topical administration. Such advantageous properties include: [0140] Species cross-reactivity. This is a typical requirement for preclinical pharmacodynamics and pharmacokinetic evaluation; [0141] Protease stability. Bicyclic peptide ligands should in most circumstances demonstrate stability to plasma proteases, epithelial (“membrane-anchored”) proteases, gastric and intestinal proteases, lung surface proteases, intracellular proteases and the like. Protease stability should be maintained between different species such that a bicyclic peptide lead candidate can be developed in animal models as well as administered with confidence to humans; [0142] Desirable solubility profile. This is a function of the proportion of charged and hydrophilic versus hydrophobic residues and intra/inter-molecular H-bonding, which is important for formulation and absorption purposes; and [0143] An optimal plasma half-life in the circulation.

Depending upon the clinical indication and treatment regimen, it may be required to develop a bicyclic peptide with short or prolonged in vivo exposure times for the management of either chronic or acute disease states. The optimal exposure time will be governed by the requirement for sustained exposure (for maximal therapeutic efficiency) versus the requirement for short exposure times to minimise toxicological effects arising from sustained exposure to the agent.

Pharmaceutically Acceptable Salts

[0144] It will be appreciated that salt forms are within the scope of this invention, and references to peptide ligands include the salt forms of said ligands.

[0145] The salts of the present invention can be synthesized from the parent compound that contains a basic or acidic moiety by conventional chemical methods such as methods described in *Pharmaceutical Salts: Properties, Selection, and Use*, P. Heinrich Stahl (Editor), Camille G. Wermuth (Editor), ISBN: 3-90639-026-8, Hardcover, 388 pages, August 2002. Generally, such salts can be prepared by reacting the free acid or base forms of these compounds with the appropriate base or acid in water or in an organic solvent, or in a mixture of the two.

[0146] Acid addition salts (mono- or di-salts) may be formed with a wide variety of acids, both inorganic and organic. Examples of acid addition salts include mono- or di-salts formed with an acid selected from the group consisting of acetic, 2,2-dichloroacetic, adipic, alginic, ascorbic (e.g. L-ascorbic), L-aspartic, benzenesulfonic, benzoic, 4-acetamidobenzoic, butanoic, (+) camphoric, camphor-sulfonic, (+)-(1S)-camphor-10-sulfonic, capric, caproic, caprylic, cinnamic, citric, cyclamic, dodecylsulfuric, ethane-1,2-disulfonic, ethanesulfonic, 2-hydroxyethanesulfonic, formic, fumaric, galactaric, gentisic, glucoheptonic, D-gluconic, glucuronic (e.g. D-glucuronic), glutamic (e.g. L-glutamic), α -oxoglutaric, glycolic, hippuric, hydrohalic acids (e.g. hydrobromic, hydrochloric, hydriodic), isethionic, lactic (e.g. (+)-L-lactic, ($\pm$)-DL-lactic), lactobionic, maleic, malic, (-)-L-malic, malonic, ($\pm$)-DL-mandelic, methanesulfonic, naphthalene-2-sulfonic, naphthalene-1,5-disulfonic, 1-hydroxy-2-naphthoic, nicotinic, nitric, oleic, orotic, oxalic, palmitic, pamoic, phosphoric, propionic, pyruvic, L-pyroglutamic, salicylic, 4-amino-salicylic, sebacic, stearic, succinic, sulfuric, tannic, (+)-L-tartaric, thiocyanic, p-toluenesulfonic, undecylenic and valeric acids, as well as acylated amino acids and cation exchange resins.

[0147] One particular group of salts consists of salts formed from acetic, hydrochloric, hydriodic, phosphoric, nitric, sulfuric, citric, lactic, succinic, maleic, malic, isethionic, fumaric, benzenesulfonic, toluenesulfonic, sulfuric, methanesulfonic (mesylate), ethanesulfonic, naphthalenesulfonic, valeric, propanoic, butanoic, malonic, glucuronic and lactobionic acids. One particular salt is the hydrochloride salt. Another particular salt is the acetate salt.

[0148] If the compound is anionic, or has a functional group which may be anionic (e.g. —COOH may be $\text{—COO}^{\text{sup.}-}$), then a salt may be formed with an organic or inorganic base, generating a suitable cation. Examples of suitable inorganic cations include, but are not limited to, alkali metal ions such as $\text{Li}^{\text{sup.}+}$, $\text{Na}^{\text{sup.}+}$ and $\text{K}^{\text{sup.}+}$, alkaline earth metal cations such as $\text{Ca}^{\text{sup.}2+}$ and $\text{Mg}^{\text{sup.}2+}$, and other cations such as $\text{Al}^{\text{sup.}3+}$ or $\text{Zn}^{\text{sup.}+}$. Examples of suitable organic cations include, but are not limited to, ammonium ion (i.e. $\text{NH}_4^{\text{sup.}+}$) and substituted ammonium ions (e.g. $\text{NH}_3\text{R}^{\text{sup.}+}$, $\text{NH}_2\text{R}_2\text{sup.}^+$, $\text{NHR}_3\text{sup.}^+$, $\text{NR}_4\text{sup.}^+$).

[0149] Examples of some suitable substituted ammonium ions are those derived from: methylamine, ethylamine, diethylamine, propylamine, dicyclohexylamine, triethylamine, butylamine, ethylenediamine, ethanolamine, diethanolamine, piperazine, benzylamine, phenylbenzylamine, choline, meglumine, and tromethamine, as well as amino acids, such as lysine and arginine. An example of a common quaternary ammonium ion is $\text{N}(\text{CH}_3)_4^{\text{sup.}+}$.

[0150] Where the peptides of the invention contain an amine function, these may form quaternary ammonium salts, for example by reaction with an alkylating agent according to methods well known to the skilled person. Such quaternary ammonium compounds are within the scope of the peptides of the invention.

Modified Derivatives

[0151] It will be appreciated that modified derivatives of the peptide ligands as defined herein are within the scope of the present invention. Examples of such suitable modified derivatives include one or more modifications selected from: N-terminal and/or C-terminal modifications; replacement of one or more amino acid residues with one or more non-natural amino acid residues (such as replacement of one or more polar amino acid residues with one or more isosteric or isoelectronic amino acids; replacement of one or more non-polar amino acid residues with other non-natural isosteric or isoelectronic amino acids); addition of a spacer group; replacement of one or more oxidation sensitive amino acid residues with one or more oxidation resistant amino acid residues; replacement of one or more amino acid residues with one or more replacement amino acids, such as an alanine, replacement of one or more L-amino acid residues with one or more D-amino acid residues; N-alkylation of one or more amide bonds within the bicyclic peptide ligand; replacement of one or more peptide bonds with a surrogate bond; peptide backbone length modification; substitution of the hydrogen on the α -carbon of one or more amino acid residues with another chemical group; modification of amino acids such as cysteine, lysine, glutamate/aspartate and tyrosine with suitable amine, thiol, carboxylic acid and phenol-reactive reagents so as to functionalise said amino acids; and introduction or replacement of amino acids that introduce orthogonal reactivities that are suitable for functionalisation, for example azide or alkyne-group bearing amino acids that allow functionalisation with alkyne or azide-bearing moieties,

respectively.

[0152] In one embodiment, the modified derivative comprises an N-terminal and/or C-terminal modification. In a further embodiment, wherein the modified derivative comprises an N-terminal modification using suitable amino-reactive chemistry, and/or C-terminal modification using suitable carboxy-reactive chemistry. In a further embodiment, said N-terminal or C-terminal modification comprises addition of an effector group, including but not limited to a cytotoxic agent, a radiochelator or a chromophore.

[0153] In a further embodiment, the modified derivative comprises an N-terminal modification. In a further embodiment, the N-terminal modification comprises an N-terminal acetyl group. In this embodiment, the N-terminal residue is capped with acetic anhydride or other appropriate reagents during peptide synthesis leading to a molecule which is N-terminally acetylated. This embodiment provides the advantage of removing a potential recognition point for aminopeptidases and avoids the potential for degradation of the bicyclic peptide.

[0154] In an alternative embodiment, the N-terminal modification comprises the addition of a molecular spacer group which facilitates the conjugation of effector groups and retention of potency of the bicyclic peptide to its target.

[0155] In a further embodiment, the modified derivative comprises a C-terminal modification. In a further embodiment, the C-terminal modification comprises an amide group. In this embodiment, the C-terminal residue is synthesized as an amide during peptide synthesis leading to a molecule which is C-terminally amidated. This embodiment provides the advantage of removing a potential recognition point for carboxypeptidase and reduces the potential for proteolytic degradation of the bicyclic peptide.

[0156] In one embodiment, the modified derivative comprises replacement of one or more amino acid residues with one or more non-natural amino acid residues. In this embodiment, non-natural amino acids may be selected having isosteric/isoelectronic side chains which are neither recognised by degradative proteases nor have any adverse effect upon target potency.

[0157] Alternatively, non-natural amino acids may be used having constrained amino acid side chains, such that proteolytic hydrolysis of the nearby peptide bond is conformationally and sterically impeded. In particular, these concern proline analogues, bulky sidechains, Ca-disubstituted derivatives (for example, aminoisobutyric acid, Aib), and cyclo amino acids, a simple derivative being amino-cyclopropylcarboxylic acid.

[0158] In one embodiment, the modified derivative comprises the addition of a spacer group. In a further embodiment, the modified derivative comprises the addition of a spacer group to the N-terminal cysteine and/or the C-terminal cysteine.

[0159] In one embodiment, the modified derivative comprises replacement of one or more oxidation sensitive amino acid residues with one or more oxidation resistant amino acid residues. In a further embodiment, the modified derivative comprises replacement of a tryptophan residue with a naphthylalanine or alanine residue. This embodiment provides the advantage of improving the pharmaceutical stability profile of the resultant bicyclic peptide ligand.

[0160] In one embodiment, the modified derivative comprises replacement of one or more charged amino acid residues with one or more hydrophobic amino acid residues. In an alternative embodiment, the modified derivative comprises replacement of one or more hydrophobic amino acid residues with one or more charged amino acid residues. The correct balance of charged versus hydrophobic amino acid residues is an important characteristic of the bicyclic peptide ligands. For example, hydrophobic amino acid residues influence the degree of plasma protein binding and thus the concentration of the free available fraction in plasma, while charged amino acid residues (in particular arginine) may influence the interaction of the peptide with the phospholipid membranes on cell surfaces. The two in combination may influence half-life, volume of distribution and exposure of the peptide drug, and can be tailored according to the clinical endpoint. In addition, the correct combination and number of charged versus hydrophobic amino acid residues may reduce irritation at the injection site (if the peptide drug has been administered subcutaneously).

[0161] In one embodiment, the modified derivative comprises replacement of one or more L-amino acid residues with one or more D-amino acid residues. This embodiment is believed to increase proteolytic stability by steric hindrance and by a propensity of D-amino acids to stabilise β -turn conformations (Tugyi et al. (2005) PNAS, 102(2), 413-418).

[0162] In one embodiment, the modified derivative comprises removal of any amino acid residues and substitution with alanines, such as D-alanines. This embodiment provides the advantage of identifying key binding residues and removing potential proteolytic attack site(s).

[0163] It should be noted that each of the above mentioned modifications serve to deliberately improve the potency or stability of the peptide. Further potency improvements based on modifications may be achieved through the following mechanisms: [0164] Incorporating hydrophobic moieties that exploit the hydrophobic effect and lead to lower off rates, such that higher affinities are achieved; [0165] Incorporating charged groups that exploit long-range ionic interactions, leading to faster on rates and to higher affinities (see for example Schreiber et al., *Rapid, electrostatically assisted association of proteins* (1996), Nature Struct. Biol. 3, 427-31); and [0166] Incorporating additional constraint into the peptide, by for example constraining side chains of amino acids correctly such that loss in entropy is minimal upon target binding, constraining the torsional angles of the backbone such that loss in entropy

is minimal upon target binding and introducing additional cyclisations in the molecule for identical reasons. (for reviews see Gentilucci et al., Curr. Pharmaceutical Design, (2010), 16, 3185-203, and Nestor et al., Curr. Medicinal Chem (2009), 16, 4399-418).

Isotopic Variations

[0167] The present invention includes all pharmaceutically acceptable (radio)isotope-labelled peptide ligands of the invention, wherein one or more atoms are replaced by atoms having the same atomic number, but an atomic mass or mass number different from the atomic mass or mass number usually found in nature, and peptide ligands of the invention, wherein metal chelating groups are attached (termed “effector”) that are capable of holding relevant (radio)isotopes, and peptide ligands of the invention, wherein certain functional groups are covalently replaced with relevant (radio)isotopes or isotopically labelled functional groups.

[0168] Examples of isotopes suitable for inclusion in the peptide ligands of the invention comprise isotopes of hydrogen, such as ^2H (D) and ^3H (T), carbon, such as ^{11}C , ^{13}C and ^{14}C , chlorine, such as ^{36}Cl , fluorine, such as ^{18}F , iodine, such as ^{123}I , ^{125}I and ^{131}I , nitrogen, such as ^{13}N and ^{15}N , oxygen, such as ^{15}O , ^{17}O and ^{18}O , phosphorus, such as ^{32}P , sulphur, such as S, copper, such as ^{64}Cu , gallium, such as ^{67}Ga or ^{68}Ga , yttrium, such as ^{90}Y and lutetium, such as ^{177}Lu , and Bismuth, such as ^{213}Bi .

[0169] Certain isotopically-labelled peptide ligands of the invention, for example, those incorporating a radioactive isotope, are useful in drug and/or substrate tissue distribution studies, and to clinically assess the presence and/or absence of the target on diseased tissues. The peptide ligands of the invention can further have valuable diagnostic properties in that they can be used for detecting or identifying the formation of a complex between a labelled compound and other molecules, peptides, proteins, enzymes or receptors. The detecting or identifying methods can use compounds that are labelled with labelling agents such as radioisotopes, enzymes, fluorescent substances, luminous substances (for example, luminol, luminol derivatives, luciferin, aequorin and luciferase), etc. The radioactive isotopes tritium, i.e. ^3H (T), and carbon-14, i.e. ^{14}C , are particularly useful for this purpose in view of their ease of incorporation and ready means of detection.

[0170] Substitution with heavier isotopes such as deuterium, i.e. ^2H (D), may afford certain therapeutic advantages resulting from greater metabolic stability, for example, increased in vivo half-life or reduced dosage requirements, and hence may be preferred in some circumstances.

[0171] Substitution with positron emitting isotopes, such as ^{11}C , ^{18}F , ^{10}O and ^{13}N , can be useful in Positron Emission Topography (PET) studies for examining target occupancy.

[0172] Isotopically-labelled compounds of peptide ligands of the invention can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described in the accompanying Examples using an appropriate isotopically-labelled reagent in place of the non-labelled reagent previously employed.

Molecular Scaffold

[0173] In one embodiment, the molecular scaffold comprises a non-aromatic molecular scaffold. References herein to “non-aromatic molecular scaffold” refers to any molecular scaffold as defined herein which does not contain an aromatic (i.e. unsaturated) carbocyclic or heterocyclic ring system.

[0174] Suitable examples of non-aromatic molecular scaffolds are described in Heinis et al. (2014) Angewandte Chemie, International Edition 53(6) 1602-1606.

[0175] As noted in the foregoing documents, the molecular scaffold may be a small molecule, such as a small organic molecule.

[0176] In one embodiment the molecular scaffold may be a macromolecule. In one embodiment the molecular scaffold is a macromolecule composed of amino acids, nucleotides or carbohydrates.

[0177] In one embodiment the molecular scaffold comprises reactive groups that are capable of reacting with functional group(s) of the polypeptide to form covalent bonds. The molecular scaffold may comprise chemical groups which form the linkage with a peptide, such as amines, thiols, alcohols, ketones, aldehydes, nitriles, carboxylic acids, esters, alkenes, alkynes, azides, anhydrides, succinimides, maleimides, alkyl halides and acyl halides.

[0178] In one embodiment, the molecular scaffold is 1,1',1''-(1,3,5-triazine-1,3,5-triyl)triprop-2-en-1-one (also known as triacryloylhexahydro-s-triazine (TATA):

##STR00002##

[0179] Thus, following cyclisation with the bicyclic peptides of the invention on the three cysteine residues, the molecular scaffold forms a tri-substituted 1,1',1''-(1,3,5-triazine-1,3,5-triyl)tripropan-1-one derivative of TATA having the following structure:

##STR00003##

wherein * denotes the point of attachment of the three cysteine residues.

[0180] In an alternative embodiment, the molecular scaffold is 1,3,5-tris(bromoacetyl) hexahydro-1, 3,5-triazine (TATB):

##STR00004##

[0181] Thus, following cyclisation with the bicyclic peptides of the invention on the cysteine residues, the molecular scaffold forms a tri-substituted 1,3,5-tris(bromoacetyl) hexahydro-1,3,5-triazine derivative of TATB having the following structure:

##STR00005##

wherein * denotes the point of attachment of the three cysteine residues.

Synthesis

[0182] The peptides of the present invention may be manufactured synthetically by standard techniques followed by reaction with a molecular scaffold in vitro. When this is performed, standard chemistry may be used. This enables the rapid large scale preparation of soluble material for further downstream experiments or validation. Such methods could be accomplished using conventional chemistry such as that disclosed in Timmerman et al. (supra).

[0183] Thus, the invention also relates to the manufacture of polypeptides or conjugates selected as set out herein, wherein the manufacture comprises optional further steps as explained below. In one embodiment, these steps are carried out on the end product polypeptide/conjugate made by chemical synthesis.

[0184] Optionally amino acid residues in the polypeptide of interest may be substituted when manufacturing a conjugate or complex.

[0185] Peptides can also be extended, to incorporate for example another loop and therefore introduce multiple specificities.

[0186] To extend the peptide, it may simply be extended chemically at its N-terminus or C-terminus or within the loops using orthogonally protected lysines (and analogues) using standard solid phase or solution phase chemistry. Standard (bio)conjugation techniques may be used to introduce an activated or activatable N- or C-terminus.

Alternatively, additions may be made by fragment condensation or native chemical ligation e.g. as described in (Dawson et al. 1994. Synthesis of Proteins by Native Chemical Ligation. Science 266:776-779), or by enzymes, for example using subtiligase as described in (Chang et al. Proc Natl Acad Sci USA. 1994 Dec. 20; 91(26):12544-8 or in Hikari et al. Bioorganic & Medicinal Chemistry Letters Volume 18, Issue 22, 15 Nov. 2008, Pages 6000-6003).

[0187] Alternatively, the peptides may be extended or modified by further conjugation through disulphide bonds. This has the additional advantage of allowing the first and second peptide to dissociate from each other once within the reducing environment of the cell. In this case, the molecular scaffold (e.g. TATA or TATB) could be added during the chemical synthesis of the first peptide so as to react with the three cysteine groups; a further cysteine or thiol could then be appended to the N- or C-terminus of the first peptide, so that this cysteine or thiol only reacted with a free cysteine or thiol of the second peptide, forming a disulphide-linked bicyclic peptide-peptide conjugate.

[0188] Furthermore, addition of other functional groups or effector groups may be accomplished in the same manner, using appropriate chemistry, coupling at the N- or C-termini or via side chains. In one embodiment, the coupling is conducted in such a manner that it does not block the activity of either entity.

Pharmaceutical Compositions

[0189] According to a further aspect of the invention, there is provided a pharmaceutical composition comprising a peptide ligand as defined herein in combination with one or more pharmaceutically acceptable excipients.

[0190] Generally, the present peptide ligands will be utilised in purified form together with pharmacologically appropriate excipients or carriers. Typically, these excipients or carriers include aqueous or alcoholic/aqueous solutions, emulsions or suspensions, including saline and/or buffered media. Parenteral vehicles include sodium chloride solution, Ringer's dextrose, dextrose and sodium chloride and lactated Ringer's. Suitable physiologically-acceptable adjuvants, if necessary to keep a polypeptide complex in suspension, may be chosen from thickeners such as carboxymethylcellulose, polyvinylpyrrolidone, gelatin and alginates.

[0191] Intravenous vehicles include fluid and nutrient replenishers and electrolyte replenishers, such as those based on Ringer's dextrose. Preservatives and other additives, such as antimicrobials, antioxidants, chelating agents and inert gases, may also be present (Mack (1982) Remington's Pharmaceutical Sciences, 16th Edition).

[0192] The peptide ligands of the present invention may be used as separately administered compositions or in conjunction with other agents. These can include antibodies, antibody fragments and various immunotherapeutic drugs, such as cyclosporine, methotrexate, adriamycin or cisplatin and immunotoxins. Further examples of other agents which may be administered separately or in conjunction with the peptide ligands of the invention include cytokines, lymphokines, other hematopoietic factors, thrombolytic and anti-thrombotic factors. Pharmaceutical compositions can include "cocktails" of various cytotoxic or other agents in conjunction with the protein ligands of the present invention, or even combinations of selected polypeptides according to the present invention having different specificities, such as polypeptides selected using different target ligands, whether or not they are pooled prior to administration.

[0193] The route of administration of pharmaceutical compositions according to the invention may be any of those commonly known to those of ordinary skill in the art. For therapy, the peptide ligands of the invention can be administered to any patient in accordance with standard techniques. The administration can be by any appropriate mode, including parenterally, intravenously, intramuscularly, intraperitoneally, transdermally, via the pulmonary route, or also, appropriately, by direct infusion with a catheter. Preferably, the pharmaceutical compositions according

to the invention will be administered intravenously. The dosage and frequency of administration will depend on the age, sex and condition of the patient, concurrent administration of other drugs, counterindications and other parameters to be taken into account by the clinician.

[0194] The peptide ligands of this invention can be lyophilised for storage and reconstituted in a suitable carrier prior to use. This technique has been shown to be effective and art-known lyophilisation and reconstitution techniques can be employed. It will be appreciated by those skilled in the art that lyophilisation and reconstitution can lead to varying degrees of activity loss and that levels may have to be adjusted upward to compensate.

[0195] The compositions containing the present peptide ligands or a cocktail thereof can be administered for prophylactic and/or therapeutic treatments. In certain therapeutic applications, an adequate amount to accomplish at least partial inhibition, suppression, modulation, killing, or some other measurable parameter, of a population of selected cells is defined as a “therapeutically-effective dose”. Amounts needed to achieve this dosage will depend upon the severity of the disease, but generally range from 0.005 to 5.0 mg of selected peptide ligand per kilogram of body weight, with doses of 0.05 to 2.0 mg/kg/dose being more commonly used. For prophylactic applications, compositions containing the present peptide ligands or cocktails thereof may also be administered in similar or slightly lower dosages.

[0196] A composition containing a peptide ligand according to the present invention may be utilised in prophylactic and therapeutic settings to aid in the alteration, inactivation, killing or removal of a select target cell population in a mammal. In addition, the peptide ligands described herein may be used extracorporeally or in vitro selectively to kill, deplete or otherwise effectively remove a target cell population from a heterogeneous collection of cells. Blood from a mammal may be combined extracorporeally with the selected peptide ligands whereby the undesired cells are killed or otherwise removed from the blood for return to the mammal in accordance with standard techniques.

Therapeutic Uses

[0197] The bicyclic peptides of the invention have specific utility as transferrin receptor 1 (TfR1) binding agents. According to a further aspect of the invention, there is provided a peptide ligand or pharmaceutical composition as defined herein for use in preventing, suppressing or treating a disease or disorder through TfR1 mediated delivery of a therapeutic agent.

[0198] Transferrins are glycoproteins found in vertebrates which bind to and consequently mediate the transport of Iron (Fe) through blood plasma. It is produced in the liver and contains binding sites for two Fe^{3+} atoms. Human transferrin is encoded by the TF gene and produced as a 76 kDa glycoprotein.

[0199] Transferrin glycoproteins bind iron tightly, but reversibly. Although iron bound to transferrin is less than 0.1% (4 mg) of total body iron, it forms the most vital iron pool with the highest rate of turnover (25 mg/24 h). Transferrin has a molecular weight of around 80 kDa and contains two specific high-affinity Fe(III) binding sites. The affinity of transferrin for Fe(III) is extremely high (association constant is 10^{20} M^{-1} at pH 7.4) but decreases progressively with decreasing pH below neutrality. Transferrins are not limited to only binding to iron but also to different metal ions. These glycoproteins are located in various bodily fluids of vertebrates. When not bound to iron, transferrin is known as “apotransferrin”.

[0200] In one embodiment, the transferrin is mammalian transferrin. In a further embodiment, the mammalian transferrin is human transferrin. In one embodiment, the human transferrin is human transferrin receptor 1 (TfR1; also known as CD71).

[0201] It will be appreciated that TfR1 binding peptides may be useful in the treatment of neurological disorders. Examples of such neurological disorders include but are not limited to: a neuropathy disorder, a neurodegenerative disease, cancer, an ocular disease disorder, a seizure disorder, a lysosomal storage disease, amyloidosis, a viral or microbial disease, ischemia, a behavioural disorder, and CNS inflammation.

[0202] In one embodiment, the neurological disorder is in a human subject. It will be appreciated that the dose amount and/or frequency of administration is modulated to reduce the concentration of peptide ligand to which the red blood cells are exposed. In a further embodiment, the treatment further comprises the step of monitoring the human subject for depletion of red blood cells.

[0203] References herein to the term “prevention” involves administration of the protective composition prior to the induction of the disease. “Suppression” refers to administration of the composition after an inductive event, but prior to the clinical appearance of the disease. “Treatment” involves administration of the protective composition after disease symptoms become manifest.

[0204] Animal model systems which can be used to screen the effectiveness of the peptide ligands in protecting against or treating the disease are available. The use of animal model systems is facilitated by the present invention, which allows the development of polypeptide ligands which can cross react with human and animal targets, to allow the use of animal models.

[0205] Transferrin receptor 1 (TfR1) is an extensively studied model receptor-ligand system and has provided considerable insight into the cellular properties and mechanisms of nutrient/scavenger receptor cargo internalization and endocytic sorting (Qian et al (2002) Pharmacological Reviews 54(4), 561-587). TfR1 is known to undergo constitutive endocytosis and recycling to the plasma membrane and possesses pH-dependent ligand binding to enable

proper sorting of endocytosed cargo. Anti-TfR1 antibodies have previously been believed to be the primary agents for TfR1 targeting of oligonucleotide therapeutics, however, the present TfR1 binding peptide ligands of the invention have the potential for demonstrating efficient and profound knockdown of gene expression in skeletal and cardiac muscle via systemically delivered TfR1-Bicyclic Peptide-siRNA conjugates.

[0206] Thus, in light of this mechanism it is believed that the peptide ligands of the invention may find utility as tissue delivery complexes, such as delivery of the TfR1-peptide ligand-payload (i.e. siRNA) complex to tissue cells, in particular muscle cells.

[0207] Thus, according to a further aspect of the invention there is provided a tissue delivery complex which comprises a peptide ligand of the invention bound to TfR1 in combination with a payload, such as another peptide, small molecule drug or oligonucleotide, in particular siRNA.

[0208] Said tissue delivery complexes therefore find utility in the treatment of musculoskeletal disorders. Examples of suitable musculoskeletal disorders include, but are not limited, to: [0209] 12q14 microdeletion syndrome [0210] 2q37 deletion syndrome [0211] 3M syndrome [0212] Absence of Tibia [0213] Absence of tibia with polydactyly [0214] Absent patella [0215] Acheiropody [0216] Achondrogenesis type 1A—See Achondrogenesis [0217] Achondrogenesis type 1B—See Achondrogenesis [0218] Achondrogenesis type 2—See Achondrogenesis [0219] Achondroplasia [0220] Acro-pectoro-renal field defect [0221] Acrocallosal syndrome, Schinzel type [0222] Acrocapitofemoral dysplasia [0223] Acrocephalopolydactyly [0224] Acrodysostosis [0225] Acrodysplasia scoliosis [0226] Acrofacial dysostosis Catania type [0227] Acrofacial dysostosis Palagonia type [0228] Acrofacial dysostosis Rodriguez type [0229] Acrofrontofacionasal dysostosis syndrome [0230] Acromelic frontonasal dysostosis [0231] Acromesomelic dysplasia [0232] Acromesomelic dysplasia Hunter Thompson type [0233] Acromesomelic dysplasia Maroteaux type [0234] Acromicric dysplasia [0235] Acroosteolysis dominant type [0236] Acropectoral syndrome [0237] Acropectorovertebral dysplasia F form [0238] Acute febrile neutrophilic dermatosis [0239] Adactylia unilateral [0240] Adams-Oliver syndrome [0241] Adenosine Deaminase 2 deficiency [0242] ADULT syndrome [0243] Adult-onset Still's disease [0244] Aicardi-Goutieres syndrome [0245] Al Gazali Sabrinathan Nair syndrome [0246] Allain-Babin-Demarquez syndrome [0247] Alpha-mannosidosis [0248] Amyotrophy, neurogenic scapuloperoneal, New England type [0249] Anauxetic dysplasia [0250] Angel shaped phalangoepiphyseal dysplasia [0251] Ankyloblepharon-ectodermal defects-cleft lip/palate syndrome [0252] Ankylosing spondylitis—Not a rare disease [0253] Ankylosing vertebral hyperostosis with tylosis [0254] Anonychia-onychodystrophy with hypoplasia or absence of distal phalanges [0255] Antley Bixler syndrome [0256] Apert syndrome [0257] Arthrogryposis multiplex congenita [0258] Arts syndrome [0259] Aspartylglycosaminuria [0260] Atelosteogenesis type 1 [0261] Atelosteogenesis type 2 [0262] Atelosteogenesis type 3 [0263] Auralcephalosyndactyly [0264] Auriculo-condylar syndrome [0265] Auriculoosteodysplasia [0266] Autosomal dominant spondyloepiphyseal dysplasia tarda [0267] Autosomal recessive early-onset inflammatory bowel disease [0268] Autosomal recessive protein C deficiency [0269] Axial osteomalacia [0270] Axial spondylometaphyseal dysplasia [0271] Baby rattle pelvic dysplasia [0272] Baller-Gerold syndrome [0273] Banki syndrome [0274] Beare-Stevenson cutis gyrata syndrome [0275] Behget disease [0276] Benallegue Lacete syndrome [0277] Bethlehem myopathy [0278] Beukes familial hip dysplasia [0279] Blau syndrome [0280] Blount disease [0281] BOD syndrome [0282] Bone dysplasia Azouz type [0283] Bone dysplasia lethal Holmgren type [0284] Boomerang dysplasia [0285] Bowing of legs, anterior with dwarfism [0286] Brachycephalofrontonasal dysplasia [0287] Brachydactylous dwarfism Mseleni type [0288] Brachydactyly elbow wrist dysplasia [0289] Brachydactyly long thumb type [0290] Brachydactyly Mononen type [0291] Brachydactyly type A1 [0292] Brachydactyly type A2 [0293] Brachydactyly type A4 [0294] Brachydactyly type A5 [0295] Brachydactyly type A6 [0296] Brachydactyly type A7 [0297] Brachydactyly type B [0298] Brachydactyly type C [0299] Brachydactyly type E [0300] Brachydactyly types B and E combined [0301] Brachyolmia type 3 [0302] Branchial arch syndrome X-linked [0303] Brody myopathy [0304] Bruck syndrome 1 [0305] Buschke-Ollendorff syndrome [0306] C syndrome [0307] Caffey disease [0308] Campomelia Cumming type [0309] Campomelic dysplasia [0310] Camptobrachydactyly [0311] Camptodactyly arthropathy coxa vara pericarditis syndrome [0312] Camptodactyly syndrome Guadalajara type 2 [0313] Camptodactyly, tall stature, and hearing loss syndrome [0314] Camurati-Engelmann disease [0315] Cantu syndrome [0316] Carpenter syndrome [0317] Carpotarsal osteochondromatosis [0318] 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to cytochrome P450 oxidoreductase deficiency [0350] Congenital contractural arachnoid [0351] Congenital femoral deficiency [0352] Congenital primary aphakia [0353] Congenital radioulnar synostosis [0354] Cornelia de Lange syndrome [0355] Cousin syndrome [0356] Craniodiaphyseal dysplasia [0357] Cranioectodermal dysplasia [0358] Craniofacial dysostosis with diaphyseal hyperplasia [0359] Craniofacial dyssynostosis [0360] Craniofrontonasal dysplasia [0361] Craniometaphyseal dysplasia, autosomal dominant [0362] Craniometaphyseal dysplasia, autosomal recessive type [0363] Craniosynostosis, anal anomalies, and porokeratosis [0364] Craniotelencephalic dysplasia [0365] Crouzon syndrome [0366] Culler-Jones syndrome [0367] Currarino triad [0368] Curry Jones syndrome [0369] Czech dysplasia metatarsal type [0370] Dandy-Walker malformation with postaxial polydactyly [0371] Dandy-Walker malformation with sagittal craniosynostosis and hydrocephalus [0372] Deficiency of interleukin-1 receptor 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dysplasia short limb-hand type [0777] Spondylometaphyseal dysplasia Algerian type [0778] Spondylometaphyseal dysplasia corner fracture type [0779] Spondylometaphyseal dysplasia Sedaghatian type [0780] Spondylometaphyseal dysplasia type A4 [0781] Spondylometaphyseal dysplasia with cone-rod dystrophy [0782] Spondylometaphyseal dysplasia with dentinogenesis imperfecta [0783] Spondylometaphyseal dysplasia X-linked [0784] Spondylometaphyseal dysplasia, Kozlowski type [0785] Spondyloperipheral dysplasia [0786] Spondylothoracic dysostosis [0787] Sprengel deformity [0788] STAR syndrome [0789] Stiff person syndrome [0790] Stuve-Wiedemann syndrome [0791] Symphalangism with multiple anomalies of hands and feet [0792] Syndactyly Cenani Lenz type [0793] Syndactyly type 3 [0794] Syndactyly type 5 [0795] Syndactyly type 9 [0796] Syndactyly-polydactyly-earlobe syndrome [0797] Syngnathia multiple anomalies [0798] Synovial Chondromatosis [0799] Systemic onset juvenile idiopathic arthritis [0800] TAR syndrome [0801] TARP syndrome [0802] Tarsal carpal coalition syndrome [0803] Tarsal tunnel syndrome [0804] Tetra-amelia syndrome [0805] Tetraamelia-multiple malformations syndrome [0806] Tetramelic monodactyly [0807] Thanatophoric dysplasia type 1 [0808] Thanatophoric dysplasia type 2 [0809] Thoracic dysplasia hydrocephalus syndrome [0810] Thoracalaryngopelvic dysplasia [0811] Tibia absent polydactyly arachnoid cyst [0812] Tietze syndrome [0813] TMEM165-CDG (CDG-Ilk) [0814] Townes-Brocks syndrome [0815] Treacher Collins syndrome [0816] Trichodonto-osseous syndrome [0817] Trichohepatoenteric syndrome [0818] Trichorhinophalangeal syndrome type 1 [0819] Trichorhinophalangeal syndrome type 2 [0820] Trichorhinophalangeal syndrome type 3 [0821] Trigonobrachycephaly, bulbous bifid nose, micrognathia, and abnormalities of the hands and feet [0822] Triphalangeal thumbs brachyectrodactyly [0823] Trochlea of the humerus aplasia of [0824] Trochlear dysplasia [0825] Troyer syndrome [0826] Tubular aggregate myopathy [0827] Tumor necrosis factor receptor-associated periodic syndrome [0828] Ulna and fibula, hypoplasia of [0829] Ulna hypoplasia-intellectual disability syndrome [0830] Ulna metaphyseal dysplasia syndrome [0831] Ulnar hypoplasia lobster claw deformity of feet [0832] Ulnar-

mammary syndrome [0833] Undifferentiated pleomorphic sarcoma [0834] Upington disease [0835] Verloes Bourguignon syndrome [0836] Viljoen Kallis Voges syndrome [0837] Warman Mulliken Hayward syndrome [0838] Weaver syndrome [0839] Weill-Marchesani syndrome [0840] Weissenbacher-Zweymuller syndrome [0841] Weyers acrofacial dysostosis [0842] Wildervanck syndrome [0843] Worth type autosomal dominant osteosclerosis [0844] Wrinkly skin syndrome [0845] X-linked dominant chondrodysplasia *punctata* 2 [0846] X-linked dominant scapuloperoneal myopathy [0847] X-linked hypophosphatemia [0848] X-linked intellectual disability-plagiocephaly syndrome [0849] X-linked skeletal dysplasia-intellectual disability syndrome [0850] Yunis-Varon syndrome [0851] The invention is further described below with reference to the following examples.

Examples

Materials and Methods

Preparation of Bicyclic Peptide Ligands (General Method)

[0852] Bicycle peptides were synthesized on Rink amide resin using standard Fmoc (9-fluorenylmethyloxycarbonyl) solid-phase peptide synthesis, either by manual coupling (for large scale) or using a Biotage Syroll automated peptide synthesizer (for small scale). Following TFA-based cleavage from the resin, peptides were precipitated with diethyl ether and dissolved in 50:50 acetonitrile/water. The crude peptides (at ~1 mM concentration) were then cyclized with 1.3 equiv. of the scaffold, using ammonium bicarbonate (100 mM) as a base. Completion of cyclization was determined by matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) or LC-MS. Once complete, the cyclization reaction was quenched using N-acetyl cysteine (10 equiv. with respect to the peptide), and the solutions were lyophilized. The residue was dissolved in an appropriate solvent and purified by RP-HPLC. Peptide fractions of sufficient purity and the correct molecular weight (verified by either MALDI-TOF and HPLC or LC-MS) were pooled and lyophilized. Concentrations were determined by UV absorption using the extinction coefficient at 280 nm, which was based on Trp/Tyr content.

[0853] All amino acids, unless noted otherwise, were used in the L-configurations.

Biological Data

[0854] The bicyclic peptide ligands of the invention were tested in the following assay:

1. TfR1 SPR Binding Assay

[0855] Biacore experiments may be performed to determine $k_{\text{sub.a}}$ ($M_{\text{sup.}}-1s_{\text{sup.}}-1$), $k_{\text{sub.d}}$ ($s_{\text{sup.}}-1$), $K_{\text{sub.D}}$ (nM) values of various peptides binding to TfR1.

[0856] Recombinant human and cynomolgus TfR1 were received from Bicycle as His-tagged TfR1 (a.a. 89-760) (ACRO Biosystems, CD1-H5243 and TFR-C524a).

[0857] For analysis of TfR1 peptide binding, a Biacore T200 or S200 instrument was used utilising a capture/coupling approach with a Cytiva NTA chip at 25° C. with 25 mM HEPES, 0.1M NaCl, 0.05% Tween 20 pH 7.4 as the running buffer. Immobilisation was carried out as follows. The chip was pre-equilibrated with an injection of 500 mM EDTA (pH 8), before activation with 5 mM NiSO₄. The surface was then activated using standard amine-coupling chemistry. Briefly, the carboxymethyl dextran surface was activated with a 1:1 ratio of 0.4 M 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC)/0.1 M N-hydroxy succinimide (NHS). The TfR1 protein (human or cynomolgus) was then captured onto the activated surface after dilution into running buffer to 200 nM and 250 nM respectively. Residual activated groups were blocked with a 7 min injection of 1 M ethanolamine (pH 8.5):HBS-N(1:1). Reference surfaces were activated and blocked as above with no TfR1 protein capture. Capture levels were in the range of 1,500-5,000 RU dependent upon the individual study Buffer was changed to 25 mM HEPES, 0.1M NaCl, 0.05% Tween 20 pH 7.4 1% DMSO.

[0858] A dilution series of test peptides was prepared in this buffer with a top peptide concentration of 5 µM and 6 further 2-fold dilutions. The SPR analysis was run at 25° C. at a flow rate of 30 µl/min with 160 seconds association and 700-800 seconds dissociation. Data were corrected for DMSO excluded volume effects. All data were double-referenced for blank injections and reference surface using standard processing procedures and data processing and kinetic fitting were performed using Scrubber software, version 2.0c (BioLogic Software). Data were fitted using simple 1:1 binding model allowing for mass transport effects where appropriate.

TABLE-US-00003
TABLE 1 Results of Human TfR1 SPR Binding Assay
BCY Number SPR Human TfR1 KD (nM)
BCY23180 50 BCY23182 22 BCY23184 13 BCY23185 12 BCY23186 27 BCY23187 38 BCY23188 109
BCY23200 9.5 BCY23201 17 BCY23202 23 BCY23203 283 BCY23204 9 BCY23205 11 BCY23206 10 BCY23207 7.7
BCY23208 6 BCY23209 5.6 BCY23210 8.6 BCY23211 7.7 BCY23216 158 BCY23217 117 BCY23218 55
BCY23219 51 BCY23220 72 BCY23221 120 BCY23222 58 BCY23223 50 BCY23224 1140 BCY23230 15
BCY23231 25 BCY23515 85

Claims

1. A peptide ligand specific for transferrin receptor 1 (TfR1) which comprises an amino acid sequence which is selected from: C[HyP][HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 1, herein referred to as BCY23180); C[Cis-HyP][HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 3, herein referred to as BCY23182); CP[Cis-

HyP]DAYLGC[tBuGly]SYCEPW (SEQ ID NO: 5, herein referred to as BCY23184);
 CP[HyP]DA[DOPA]LGC[tBuGly]SYCEPW (SEQ ID NO: 6, herein referred to as BCY23185);
 CP[HyP]DA[pCaPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 7, herein referred to as BCY23186);
 CP[HyP]DA[pCoPhe]LGC[tBuGly]SYCEPW (SEQ ID NO: 8, herein referred to as BCY23187);
 CP[HyP]DA[hTyr]LGC[tBuGly]SYCEPW (SEQ ID NO: 9, herein referred to as BCY23188);
 CP[HyP]DAYLGC[tBuGly]S[DOPA]CEPW (SEQ ID NO: 21, herein referred to as BCY23200);
 CP[HyP]DAYLGC[tBuGly]S[pCaPhe]CEPW (SEQ ID NO: 22, herein referred to as BCY23201);
 CP[HyP]DAYLGC[tBuGly]S[pCoPhe]CEPW (SEQ ID NO: 23, herein referred to as BCY23202);
 CP[HyP]DAYLGC[tBuGly]S[hTyr]CEPW (SEQ ID NO: 24, herein referred to as BCY23203);
 CP[HyP]DAYLGC[tBuGly]SYCE[HyP]W (SEQ ID NO: 25, herein referred to as BCY23204);
 CP[HyP]DAYLGC[tBuGly]SYCE[Oxa]W (SEQ ID NO: 26, herein referred to as BCY23205);
 CP[HyP]DAYLGC[tBuGly]SYCE[Cis-HyP]W (SEQ ID NO: 27, herein referred to as BCY23206);
 CP[HyP]DAYLGC[tBuGly]SYCEPY (SEQ ID NO: 28, herein referred to as BCY23207);
 CP[HyP]DAYLGC[tBuGly]SYCEP[DOPA](SEQ ID NO: 29, herein referred to as BCY23208);
 CP[HyP]DAYLGC[tBuGly]SYCEP[pCaPhe](SEQ ID NO: 30, herein referred to as BCY23209);
 CP[HyP]DAYLGC[tBuGly]SYCEP[pCoPhe](SEQ ID NO: 31, herein referred to as BCY23210);
 CP[HyP]DAYLGC[tBuGly]SYCEP[hTyr](SEQ ID NO: 32, herein referred to as BCY23211);
 CP[HyP]EAYLGC[tBuGly]SYCEPW (SEQ ID NO: 33, herein referred to as BCY23216); CP[HyP]
 [Gla]AYLGC[tBuGly]SYCEPW (SEQ ID NO: 34, herein referred to as BCY23217);
 CP[HyP]DAYSGC[tBuGly]SYCEPW (SEQ ID NO: 35, herein referred to as BCY23218);
 CP[HyP]DAYTGC[tBuGly]SYCEPW (SEQ ID NO: 36, herein referred to as BCY23219);
 CP[HyP]DAYDGC[tBuGly]SYCEPW (SEQ ID NO: 37, herein referred to as BCY23220);
 CP[HyP]DAYEGC[tBuGly]SYCEPW (SEQ ID NO: 38, herein referred to as BCY23221);
 CP[HyP]DAYNGC[tBuGly]SYCEPW (SEQ ID NO: 39, herein referred to as BCY23222);
 CP[HyP]DAYQGC[tBuGly]SYCEPW (SEQ ID NO: 40, herein referred to as BCY23223);
 CP[HyP]DAYLGC[tBuGly][HSer]YCEPW (SEQ ID NO: 41, herein referred to as BCY23224);
 CP[HyP]DAYLGC[tBuGly]SYCDPW (SEQ ID NO: 47, herein referred to as BCY23230);
 CP[HyP]DAYLGC[tBuGly]SYC[Gla]PW (SEQ ID NO: 48, herein referred to as BCY23231); and
 CP[HyP]DAYLGC[3HyV]SYCEPW (SEQ ID NO: 50, herein referred to as BCY23515), or a pharmaceutically acceptable salt thereof, wherein Cis-HyP represents cis-L-4-hydroxyproline, DOPA represents 3,4-dihydroxy-phenylalanine, Gla represents L-γ-carboxyglutamic acid, HyP represents hydroxyproline, HSer represents homoserine, hTyr represents homo-tyrosine, 3HyV represents 3-hydroxy-L-valine, Oxa represents oxazolidine-4-carboxylic acid, pCaPhe represents L-4-carbamoylphenylalanine, pCoPhe represents 4-carboxy-L-phenylalanine, tBuGly represents t-butyl-glycine.

2. The peptide ligand as defined in claim 1, which comprises an N-terminal acetyl group and a C-terminal CONH.sub.2 group.

3. The peptide ligand as defined in claim 1 or claim 2, wherein the pharmaceutically acceptable salt is selected from the free acid or the sodium, potassium, calcium or ammonium salt.

4. A bicyclic peptide ligand which comprises a peptide ligand as defined in any one of claims 1 to 3, wherein the first, second and third cysteine residues within said peptide ligands are covalently bonded to a molecular scaffold such that two polypeptide loops are formed on said molecular scaffold.

5. The bicyclic peptide ligand as defined in claim 4, wherein the molecular scaffold is a derivative of TATB which has the following structure: ##STR00006## wherein * denotes the point of attachment of the three cysteine residues.

6. A pharmaceutical composition which comprises the peptide ligand as defined in any one of claims 1 to 3 or the bicyclic peptide ligand as defined in claim 4 or claim 5, in combination with one or more pharmaceutically acceptable excipients.

7. The peptide ligand as defined in any one of claims 1 to 3, or the bicyclic peptide ligand as defined in claim 4 or claim 5, or the pharmaceutical composition of claim 6, for use in preventing, suppressing or treating a disease or disorder through Tfr1 mediated delivery of a therapeutic agent.

8. A tissue delivery complex which comprises a peptide ligand as defined in any one of claims 1 to 3 or the bicyclic peptide ligand as defined in claim 4 or claim 5, bound to Tfr1 in combination with a payload, such as an oligonucleotide, in particular siRNA.

9. The tissue delivery complex as defined in claim 8, which is a muscle tissue delivery complex.

10. The tissue delivery complex as defined in claim 8 or claim 9, for use in the treatment of a musculoskeletal disorder.